Cell Biology

Sem 1 | Watermarked study resource | Generated from local SRM dataset

===== SOURCE_FILE: nexus/Cell Biology/PYQs/PYQ Dec 2022.pdf =====
CATEGORY: PYQs
STATUS: ok

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Page 1 of 1 0 8 D F 3 - 1 8 B T C 102J

Reg. No.

B.Tech. DEGREE EXAMINATION, DECEMBER 2022
Third Semester

18BTC102J – CELL BIOLOGY
(For the candidates admitted from the academic year 2020-2021 to 2021-2022)
Note:
(i) Part - A should be answered in OMR sheet within first 40 minutes and OM R sheet should be handed
over to hall invigilator at the end of 40th minute.
(ii) Part - B should be answered in answer booklet.

Time: 2½ Hours Max. Marks: 75

PART – B (5 × 10 = 50 Marks)
Answer ALL Questions

Marks

BL

CO

PO
26. a. Discuss in detail about various molecular composition of cells.

10 3 1 1
(OR)
b. Describe the structure of cell membrane and mention its function with appropriate diagram.

10 3 1 1
27. a. Describe how lipid and polysaccharide metabolism occur i n Golgi. Add a note on protein sorting.

10 4 2 3
(OR)
b. Write a detailed note on the structure and functions of mitochondria.

10 3 2 3
28. a. Discuss about the role of actin and myosin in muscle con traction.

10 4 3 4
(OR)
b. What are the various types of cell-cell interactions?

10 3 3 4
29. a. Mention the importance of cell signaling and the pathway of intracellular signal transduction.

10 4 4 4
(OR)

b. Write a detailed essay on the stem cells and their usa ges in therapy.

10 3 4 4
30. a. Define Cancer. Discuss in detail about oral cancer along with treatment.

10 4 5 4
(OR)
b. Discuss in detail about neurodegenerative diseases wit h a note on Tau hypothesis.

10 4 6 5

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===== SOURCE_FILE: nexus/Cell Biology/PYQs/PYQ Jun 2023.pdf =====
CATEGORY: PYQs
STATUS: ok

[Page 1]
Note: (i) (ii) Time: 3 Hours
(A)
rart -A Should be answered in (OMR sheet within Grct A0 minutes and OMR sheet should be hande over to hall invigilator at the end of 40th minute. Part-B and Part -C should be answered in answer booklet.
1. Prokaryotic cell does not possess Cell wall
Page l of 4
Reg. No. B.Tech. DEGREE EXAMINATION, JUNE Z0Z3 Second Semester (For the candidates admited rom the academic year 2022-2025)
(C) Nuclear membrane
meiosis
21BTB105T -CELL BIOLOGY
2. The number of chromatids at metaphase is
PART -A (20 x l= 20Marks) Answer ALL Questions
(A) Two each in mitosis and (B) Two in mitosis and four in
(A) H,O (C) PO4
(C) Two in mitosis and one in (D) One in mitosis and two in meiosis meiosis 3. In which of the following situations would cells die by necrosis and not apoptosis?
(A) Tubulin (C) Desmin 6. Cell sap is a
(A) Removal of cells with damaged (B) Removal of heart muscle cells DNA that cannot be repaired damaged by O2 depletion
(B) Cytoplasm
(C) Removal of virus infected cells (D) Removal of developing neurons that fail to make connection with other cells 4. Which among the following was not used in Stanley Miller's original mixture?
5. Cilia and flagella are made up of
(D) Plasma membrane
meiosis
(A) Living content of the cell
(A) Chloroplast (C) Mitochondria
(B) H) (D) N: (B) Lamin (D) Keratin (B) Non-living protoplasm (C) Living content of the cytoplasm (D) Non-living vacuole 7 Which of the organelle is involved in cell wall synthesis?
Content of the content of the
(B) Golgi apparatus (D) Lysosome
Max. Marks: 75
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following cardiac infraction

E
8. Which of the following transportation mechanisms does not obey the laws of diffusion? (A) Active transport (C) Diffusion (A) G protein 9. Which of the following is a microtubule associated protein? (C) Tus protein histology? (A) Parkinson's disease 10. Which dementia is characterized by 'frontal lob' symptoms and specific (C) Pick's disease 11. An enzyme acts by (A) Increasing activation (C) Increasing the pH the energy
(C) Cytochalasin-B (A) Increases fluidity (C) Decreases fluidity
(B) Osmosis
12. Which of the following receptors activates the pathway usually by forming molecular dimers that result in protein phosphorylation reactions upon binding of their specific ligand? (A) Receptor tyrosine kinase (C) Steroid hormone receptors
(C) Golgi apparatus
(D) Facilitated diffusion
(A) 30 minutes
13. Which of the following is a microfilament inhibitor? (A) Colchicine
(C) 59 minutes
(B) Tau protein (D) Rho protein

(A) Lymphoma
(B) Vascular dementia
14. At physiological pH, increase in cholesterol level
(C) Sarcoma
(D) Frontal temporal dementia
of (B) Reducing the energy of activation
18. Leukoplakia is related to (A) Necrosis (C) Keratosis
(D) Decreasing pH
15. Which organelle is involved in xenobiotic detoxification? (A) SER
(B) Ligand-gated ion channels (D) G-protein coupled receptors (B) Cinchonine (D) Aspirin
16. A cell divides every minute. At this rate of division it can fill a 100 ml of beaker in one hour. How much time does it take to fill a 50 ml beaker?
(B) No change in fluidity (D) Alters pH (B) RER (D) Lysosome
17. Which of the following cancers do not form a solid neoplasm? (B) 60 minutes (D) 15 minutes (B) Lipoma (D) Leukemia (B) Sclerosis (D) Plaques
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1 2 43
2 4 3

FYANIN
10 Which among the following is the cause of lung cancer? (A) Cement factory (C) Calcium fluoride 0 Which of the following conditions is associated with increased risk of breast cancer? (A) Papillomatosis (C) Severe hyperplasia
(B) Coal mining (D) Bauxite mining
PART -B (5 x 8 = 40 Marks) Answer ALL Questions 21. a. What do you mean by multicellular eukaryotes? Illustrate in depth the multicellular eukaryotes that are employed in scientific experiments. (OR)
(B) Atypical hyperplasia (D) Fibrocystic mastopathy
b. Which membrane separates the interior of the cell from the outside environment? Give a through description of the membranes structural integrity. 22. a. Which cell organelle is responsible for conversion of solar energy into chemical energy for growth? Clearly describe that cell organelles structure, composition and purpose. (OR) b. Which molecules are the

second major components of membranes? How those molecules that are expressed as second major component of a membrane are transported?

23. a. What do you understand by the activation of tension -generating sites within muscle cells? How and which protein are responsible for this activation of tension-generating sites within muscle cells? (OR) b. Which junctions allow the direct chemical communication between the adjacent cytoplasm(s) in animal? Give a detailed account on the signaling between the cells through those junctions. 24. a. Which phenomenon includes a series of events that takes place in a cell as it grows and divides? Delineate the various stages of that phenomenon with particular emphasis on a process of cell duplication, in which one cell divides into two genetically identical daughter cells.

Marks

(OR) D. Which process helps to eliminate unwanted cells during early development? Explain the intrinsic and extrinsic pathways of that process.

1

(OR)

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a Y nnch cancer is the leading cause of cancer deaths world wide iTespective Ol male and female? Write an assav on etiology, pathogenesis and treatment of that particular cancer.

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[Page 4]
B.leh nr b03 203
b. Which disease occur when nerve cells in the brain or peripheral nervous system lose function over time and ultimately die? Give a detailed account on any one such diseases and its treatment. PART -C(1x 15 = 15 Marks) Answer ANY ONE Questions 26. How lipid and polysaccharide metabolism occur in Golgi? Add a note on protein sorting. 27. Identify the importance of cell signaling and the pathway of intracellular signal transduction.
Marks 15
15
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===== SOURCE_FILE: nexus/Cell Biology/PYQs/Study Material Drive Link.url =====
CATEGORY: PYQs
STATUS: url_reference_only

URL: https://drive.google.com/drive/folders/1Mgt2VtpzOxARAGFm6D2_cNl-s0PBHbAp?usp=drive_link

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Reg. No. OOo 2

B.Tech. DEGREE EXAMINATION, JUNE Z0Z3
Second Semester

21BTB105T - CELL BIOLOGY
(For the candidates admited rom the academic year 2022-2025)

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Marks BL CO PO

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(C) Nuclear membrane (D) Plasma membrane

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(A) Living content of the cell (B) Non-living Content of the
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(A) Chloroplast (B) Golgi apparatus
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Answer ALL Questions Marks BI CO PO

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(OR) 02JF2-2IBTBI0ST

B.leh nr b03 203

2,3 3

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Marks BL Co PO

Answer ANY ONE Questions
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